AI-Based Rockfall Management System for Open-Pit Mines

Day WSIE Project Tracker Report

Batch: III AIML_A – 2028

Tracking Period: June 01, 2025 – June 13, 2025

Project Duration: June 01 – June 30, 2025

■ IoT · Machine Learning · Real-Time Monitoring · Safety Automation

DAY 01

■ Planned Activity

Project Title Finalization

Jun-01 | Jun-01

■ Expected Deliverable

Approved Project Title

What I Did

On Day 1, I brainstormed and finalized the project title. I reviewed multiple AI and IoT-based mining safety ideas, evaluated technical feasibility, and selected a project that addresses a real industrial challenge using modern technologies.

Activities Breakdown

- Brainstormed 10+ project ideas focused on industrial safety and AI
- Evaluated each idea on feasibility, innovation, and academic value
- Shortlisted 3 topics: Smart Traffic AI, Hospital Management, Rockfall Prediction System
- Finalized: 'AI-Based Rockfall Management System for Open-Pit Mines'
- Got faculty approval for the title

Why This Topic?

Reason	Explanation
Real-World Impact	Rockfalls in open-pit mines cause fatalities and production loss; AI can predict and prevent them
Technology Combination	Combines IoT sensors + ML models + Real-time dashboards — novel and technically challenging
Industry Relevance	Applicable to mining companies, safety regulators, and smart infrastructure
Academic Rigor	Covers all required modules: DB, Frontend, Backend, AI, IoT, and Security

DAY 02

■ Planned Activity

Requirement Gathering

Jun-02 | Jun-02

■ Expected Deliverable

Problem Statement

Problem Statement

Problem Statement: Open-pit mines are highly vulnerable to rockfall incidents triggered by geological instability, excessive vibration, rainfall infiltration, and slope deformation. Traditional monitoring relies on manual inspections and periodic assessments, which fail to detect early warning signs. There is a need for an intelligent, automated system capable of continuously monitoring slope conditions and predicting potential rockfall hazards before they occur.

Stakeholder Analysis

Stakeholder	Requirements
Mine Workers	Real-time hazard alerts, evacuation signals, safe zone guidance
Mine Safety Officers	Dashboard for monitoring slope conditions, incident reports, historical data
Geotechnical Engineers	AI-predicted risk levels, sensor trend analysis, slope behavior data
Mine Management	Operational uptime tracking, damage prevention reports, compliance logs
System Administrator	Manage sensors, user accounts, AI model updates, system configuration

Functional Requirements

- User registration, login, and role-based profile management
- Real-time sensor data acquisition from IoT devices
- AI-based anomaly and rockfall risk prediction
- Risk level classification: Low / Medium / High
- Automated alert and notification system for high-risk conditions
- Centralized monitoring dashboard with live maps and sensor views
- Historical data analysis and trend reporting

Non-Functional Requirements

Requirement	Specification
System Uptime	99.9% availability — critical safety system
Alert Response Time	< 2 seconds from detection to notification
Data Security	End-to-end encryption + role-based JWT auth
Scalability	Support 100+ simultaneous sensor nodes and users

DAY 03

■ Planned Activity

Objective Definition

Jun-03 | Jun-03

■ Expected Deliverable

Project Objectives

Project Objectives

Obj #	Objective Description
OBJ-01	Monitor mine slope conditions in real time using IoT sensors (vibration, tilt, moisture, rainfall, crack)
OBJ-02	Collect and store environmental and geological sensor data in a structured database
OBJ-03	Detect abnormal patterns and slope instability indicators using ML algorithms
OBJ-04	Classify risk levels into Low, Medium, and High categories for mine zones
OBJ-05	Generate timely automated alerts and notifications for hazardous conditions
OBJ-06	Provide a centralized web-based monitoring dashboard for operators and engineers
OBJ-07	Support predictive maintenance and safety management through historical data analysis
OBJ-08	Reduce accidents, equipment damage, and operational downtime through early warning

SMART Goals

Criterion	Description
Specific	Build a rockfall prediction system using IoT + ML for open-pit mines in 30 days
Measurable	All 8 objectives completed with working sensor integration and ML demo
Achievable	Technologies are well-documented; I have prior Python/React/MySQL skills
Relevant	Directly addresses mine safety gaps using state-of-the-art AI
Time-Bound	Complete by June 30, 2025 with full submission and demo

DAY 04

■ Planned Activity

User & Module Identification

Jun-04 | Jun-04

■ Expected Deliverable

Module List

User Roles

User Role	Description	Access Level
Mine Worker / Operator	Receives alerts and views safety status for their zone	Basic
Safety Officer	Monitors dashboard, responds to incidents	Medium
Geotechnical Engineer	Analyzes sensor trends, manages risk assessments	Medium
System Administrator	Manages sensors, users, AI models, full system config	Full

Module List

Module #	Module Name	Description
M-01	User Management	Registration, Login, JWT, Role-based access control
M-02	Sensor Data Acquisition	Collects real-time data from IoT sensors in mining zones
M-03	Data Storage	Stores sensor readings, risk assessments, alerts, history
M-04	AI Prediction Engine	ML models (Random Forest, XGBoost) for rockfall risk prediction
M-05	Risk Assessment	Classifies zones into Low, Medium, High risk categories
M-06	Alert & Notification	Triggers warnings and notifications on anomaly detection
M-07	Dashboard & Visualization	Graphical sensor data, risk maps, alerts, and trends
M-08	Reporting Module	Generates reports for auditing and safety compliance

DAY 05

■ Planned Activity

Use Case Diagram Preparation

Jun-05 | Jun-05

■ Expected Deliverable

CRUD APIs

Use Case Summary

Actor	Use Case
Mine Worker	Login / View Safety Zones / Receive Risk Alerts
Safety Officer	Login to Dashboard / Monitor Live Sensor Data / View & Respond to Incidents
Safety Officer	Generate Incident and Risk Assessment Reports
Geotechnical Engineer	Analyze Historical Sensor Data / View AI Predictions / Manage Sensor Info
Admin	Manage Users and Roles / Configure Sensors / Update AI Model Parameters
System (AI)	Auto-detect anomalies from sensor feeds / Classify risk levels
System (Alerts)	Auto-generate warnings and send notifications to relevant personnel

CRUD API Matrix

Entity	Create	Read	Update	Delete
User	POST /api/users/register	GET /api/users/{id}	PUT /api/users/{id}	DELETE /api/users/{id}
Sensor	POST /api/sensors	GET /api/sensors/{id}	PUT /api/sensors/{id}	DELETE /api/sensors/{id}
Sensor Reading	POST /api/readings	GET /api/readings	—	—
Risk Assessment	POST /api/risk	GET /api/risk/{zone_id}	PUT /api/risk/{id}	—
Alert	POST /api/alerts	GET /api/alerts	PUT /api/alerts/{id}	—
Mine Location	POST /api/locations	GET /api/locations	PUT /api/locations/{id}	DELETE /api/locations/{id}
Report	POST /api/reports	GET /api/reports/{id}	—	—

DAY 06

■ Planned Activity

Database Requirement Analysis

Jun-06 | Jun-06

■ Expected Deliverable

Table List

Database Tables Identified

Table Name	Purpose	Key Columns
users	All user accounts and roles	user_id, name, email, role, password_hash
sensors	Registered IoT sensors in mining zones	sensor_id, type, location_id, status, installed_at
sensor_readings	Time-series sensor data	reading_id, sensor_id (FK), value, unit, timestamp
mine_locations	Mine zones and section definitions	location_id, zone_name, coordinates, risk_level
risk_assessments	AI-generated risk evaluations per zone	assessment_id, location_id (FK), risk_level, confidence, assessed_at
alerts	System-generated hazard alerts	alert_id, type, message, risk_level, sent_at, acknowledged
reports	Safety and audit reports	report_id, generated_by, type, file_path, created_at
system_logs	System activity and error logs	log_id, action, user_id, timestamp, details

Database Design Decisions

- MySQL: Used for all structured relational data (users, sensors, readings, assessments)
- All tables follow 3NF (Third Normal Form) for minimal redundancy
- Auto-increment integers used as primary keys for simplicity and performance
- Timestamps stored in UTC format for consistency across time zones
- Indexed columns: sensor_id, timestamp, risk_level for fast query performance

DAY 07

■ Planned Activity

ER Diagram Design

Jun-07 | Jun-07

■ Expected Deliverable

ER Diagram

Entity-Relationship Diagram

The ER diagram captures all entities and their relationships. Core entities and attributes are listed below.

■ USERS

Attribute	Type / Description
user_id (PK)	INT, NOT NULL, AUTO_INCREMENT
name	VARCHAR(100), NOT NULL
email	VARCHAR(150), UNIQUE, NOT NULL
password_hash	TEXT, NOT NULL
role	ENUM (worker, officer, engineer, admin)
created_at	TIMESTAMP, DEFAULT NOW()
is_active	BOOLEAN, DEFAULT TRUE

■ *One USER has many ALERTS (1:N) | One USER generates many RISK_ASSESSMENTS (1:N)*

■ SENSORS

Attribute	Type / Description
sensor_id (PK)	INT, NOT NULL, AUTO_INCREMENT
sensor_type	ENUM (vibration, tilt, moisture, rainfall, crack)
location_id (FK)	INT → mine_locations
status	ENUM (active, inactive, fault)
installed_at	TIMESTAMP

■ *One MINE_LOCATION has many SENSORS (1:N) | One SENSOR produces many READINGS (1:N)*

■ SENSOR_READINGS

Attribute	Type / Description
reading_id (PK)	INT, NOT NULL, AUTO_INCREMENT
sensor_id (FK)	INT → sensors
value	DECIMAL(10,4), NOT NULL
unit	VARCHAR(20) — e.g., Hz, °, %, mm
timestamp	TIMESTAMP, NOT NULL

■ *One SENSOR_READING may trigger one RISK_ASSESSMENT (1:0..1)*

■ RISK_ASSESSMENTS

Attribute	Type / Description
assessment_id (PK)	INT, NOT NULL, AUTO_INCREMENT
location_id (FK)	INT → mine_locations
risk_level	ENUM (low, medium, high)
confidence_score	FLOAT — 0.0 to 1.0
assessed_at	TIMESTAMP

■ *One RISK_ASSESSMENT triggers one ALERT (1:0..1)*

DAY 08

■ Planned Activity

Database Schema Creation

Jun-08 | Jun-08

■ Expected Deliverable

SQL Schema

SQL Schema Overview

The MySQL schema was created with all tables, constraints, indexes, and foreign key relationships as defined below.

Schema: Relationships Map

From Table	To Table	Relationship	Key
users	risk_assessments	1:N — user generates assessments	user_id → assessed_by
users	alerts	1:N — user receives alerts	user_id → recipient_id
mine_locations	sensors	1:N — zone has sensors	location_id → location_id
mine_locations	risk_assessments	1:N — zone has assessments	location_id → location_id
sensors	sensor_readings	1:N — sensor produces readings	sensor_id → sensor_id
risk_assessments	alerts	1:1 — assessment triggers alert	assessment_id → assessment_id

Index Strategy

Table	Index Column(s)	Reason
users	email	Fast login lookup
sensor_readings	sensor_id, timestamp	Time-series queries and sensor history
risk_assessments	location_id, risk_level	Dashboard filtering by zone and risk
alerts	risk_level, sent_at	Real-time alert sorting
mine_locations	risk_level	Quick high-risk zone queries

DAY 09

■ Planned Activity

UI Wireframe Design

Jun-09 | Jun-09

■ Expected Deliverable

Page Layouts

Wireframe Page List

Page #	Page Name	User Role	Key Elements
P-01	Login Page	All	Email/password form, role selector, login button
P-02	Main Dashboard	Officer/Engineer	Live sensor cards, risk map, active alert feed, stats
P-03	Sensor Monitoring Page	All	Sensor list, live readings, status indicators, graphs
P-04	Risk Analysis Page	Officer/Engineer	Risk level per zone, AI confidence, trend charts
P-05	Alert Management Page	Officer/Admin	Active alerts list, acknowledge button, severity filter
P-06	Historical Data Page	Engineer/Admin	Date-range picker, sensor history graphs, export
P-07	Reports Page	Admin	Report generator, download PDF, audit logs
P-08	Sensor Management Page	Admin	Add/edit/remove sensors, assign to zones, status toggle
P-09	User Management Page	Admin	User table, roles, activate/deactivate, search/filter

UI Design Principles

- Responsive design for desktop and tablet (mine control rooms and field tablets)
- Color coding: Red = High risk, Yellow = Medium risk, Green = Low risk, Blue = Info
- Dashboard-centric UI: primary view is the live risk map and sensor feed
- Real-time updates via WebSocket connection without page refresh
- React.js with Tailwind CSS for rapid, consistent UI development

DAY 10

■ Planned Activity

Login & Dashboard UI Design

Jun-10 | Jun-10

■ Expected Deliverable

UI Screens

Login Screen Design

Component	Description
Logo & Title	App logo, 'Rockfall Management System' header in dark navy
Email Input	Placeholder: 'Enter your email', format validation
Password Input	Masked input with show/hide toggle
Role Selector	Dropdown: Worker / Safety Officer / Engineer / Admin
Login Button	Primary blue button, disabled until form is valid
Forgot Password	Link below button, redirects to email reset flow
Error Messages	Red inline validation messages for invalid inputs

Main Dashboard UI Components

Component	Description
Header Bar	App title, user avatar, notification bell (with count), logout
Risk Map (Main)	Interactive map with mine zones color-coded by risk level
Sensor Stats Cards	Cards showing: Active Sensors, High-Risk Zones, Alerts Today, System Status
Live Alert Feed	Scrollable list of recent alerts with severity badges and timestamps
Sensor Reading Panel	Real-time graph showing vibration, tilt, moisture, rainfall values
Zone Risk Summary	Table of all zones with current risk level and last updated time

Safety Officer Dashboard Additional Components

- Live sensor map with zone polygons color-coded by risk level
- Incident response panel: sortable by severity, time, zone
- Statistics bar: Total sensors active, High-risk zones, Alerts triggered today
- Alert notification popup: Auto-appears for High-risk detection events

DAY 11

■ Planned Activity

Navigation & Form Design

Jun-11 | Jun-11

■ Expected Deliverable

UI Prototype

Navigation Structure

Navigation Item	Worker	Officer/Engineer	Admin
Home / Dashboard	■	■	■
Sensor Monitor	■	■	■
Risk Analysis	View only	Full Access	Full Access
Alert Management	View only	Full Access	Full Access
Historical Data	■	■	■
Sensor Management	■	View only	Full CRUD
User Management	■	■	■
Reports	■	View	■
Profile / Settings	■	■	■

Key Forms Designed

Form Name	Fields	Validation
User Registration	Name, Email, Phone, Password, Role	Email format, phone 10 digits, password strength
Sensor Registration	Sensor Type, Zone Assignment, Calibration Values, Install Date	Required fields, valid zone selection
Alert Acknowledgement	Alert ID (auto), Action Taken, Notes, Officer ID (auto-fill)	Notes required for High-risk alerts
Report Generator	Report Type, Date Range, Zone Filter, Format (PDF/CSV)	Date range must be valid, zone required
AI Config Form	Model Type, Confidence Threshold (0.5–0.99), Sensor IDs	Threshold must be 0.5–0.99 range

Prototype Flow

- Figma prototype: All 9 screens connected with click-through interactions
- Worker flow: Login → Dashboard → View Sensor Status → Receive Alert → View Details
- Officer flow: Login → Dashboard → View Alert → View Zone Risk → Acknowledge & Report
- Admin flow: Login → Admin Panel → Manage Sensors → Manage Users → Generate Report

DAY 12

■ Planned Activity

Design Review

Jun-12 | Jun-12

■ Expected Deliverable

Design Approval

Design Review Checklist

Review Area	Status	Comments
Project Title	■ Approved	Title is clear, specific, and industrially relevant
Problem Statement	■ Approved	Well-defined scope with real-world significance
Objectives (8)	■ Approved	All SMART objectives met and clearly defined
Module List (8)	■ Approved	All modules accounted for and described
Use Cases	■ Approved	Covers all actors and interactions comprehensively
CRUD APIs	■ Approved	REST conventions followed correctly
Database Tables (8)	■ Approved	3NF normalized, indexes defined
ER Diagram	■ Approved	All entity relationships correctly mapped
SQL Schema	■ Approved	Constraints and foreign keys properly defined
UI Wireframes	■ Approved	9 screens fully designed for all user roles
UI Screens	■ Approved	Login and Dashboard components complete
UI Prototype	■■ Minor Fix	Alert popup animation needs refinement for mobile
Overall Design	■ Approved	Ready to proceed to development phase

Faculty Review Feedback

1. Strong project with clear industrial safety applicability. IoT integration is well-thought-out.
2. Ensure AI module includes fallback logic for low-confidence predictions to avoid false positives causing unnecessary evacuations.
3. Alert system must have escalation levels: local alarm → supervisor → emergency services.
4. Overall: Design phase is well-executed and technically sound. Approved for development phase.

DAY 13

■ Planned Activity

Frontend Environment Setup

Jun-13 | Jun-13

■ Expected Deliverable

React Project Setup

Frontend Environment Setup

Tool / Package	Version	Purpose
Node.js	v20.x LTS	JavaScript runtime environment
npm	v10.x	Package manager
React.js	v18.2	Frontend framework
Vite	v5.x	Build tool (faster than CRA)
Tailwind CSS	v3.4	Utility-first CSS framework
React Router DOM	v6.x	Client-side routing and navigation
Axios	v1.6	HTTP client for REST API calls
Recharts / Chart.js	v3.x	Sensor data visualization and analytics
Leaflet / React-Leaflet	v4.x	Interactive mine zone maps
Socket.io-client	v4.x	Real-time WebSocket sensor updates
React Toastify	v10.x	Alert and notification toasts
ESLint + Prettier	Latest	Code quality and formatting

Project Folder Structure

```
rockfall-frontend/
■■■ src/
■   ■■■ components/   # Reusable UI components (SensorCard, RiskBadge, AlertPopup)
■   ■■■ pages/        # Route-level pages (Dashboard, SensorMonitor, RiskAnalysis)
■   ■■■ services/     # Axios API call functions (sensorService, alertService)
■   ■■■ context/      # React Context (AuthContext, AlertContext)
■   ■■■ hooks/        # Custom hooks (useSensorData, useRiskLevel)
■   ■■■ utils/        # Helper functions, validators, risk calculators
■   ■■■ App.jsx       # Root component with routes
■■■ public/           # Static assets (favicon, index.html)
■■■ package.json      # Dependencies manifest
```

Environment Variables (.env)

- VITE_API_BASE_URL=http://localhost:5000/api
- VITE_MAPS_API_KEY=[Google Maps / OpenLayers API Key]

- VITE_WS_URL=ws://localhost:5000/ws (WebSocket for real-time sensor data)
- VITE_ML_SERVICE_URL=http://localhost:8000 (Python AI microservice)

Days 1–13 Summary

Day	Date	Planned Activity	Expected Deliverable
01	Jun-01	Project Title Finalization	Approved Project Title
02	Jun-02	Requirement Gathering	Problem Statement
03	Jun-03	Objective Definition	Project Objectives
04	Jun-04	User & Module Identification	Module List
05	Jun-05	Use Case Diagram Preparation	CRUD APIs
06	Jun-06	Database Requirement Analysis	Table List
07	Jun-07	ER Diagram Design	ER Diagram
08	Jun-08	Database Schema Creation	SQL Schema
09	Jun-09	UI Wireframe Design	Page Layouts
10	Jun-10	Login & Dashboard UI Design	UI Screens
11	Jun-11	Navigation & Form Design	UI Prototype
12	Jun-12	Design Review	Design Approval
13	Jun-13	Frontend Environment Setup	React Project Setup